

MiNi Project

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Branch: MCA(AIML)

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Subject Name: Advanced Internet Programming Lab

UID: 24MCI10252

Section/Group: 24MAM-3-B

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Aim Of BankManagement System :--

The **Bank Management System** is designed to automate and streamline banking operations. Its primary objectives include:

◆ **Main Aim:**

To provide a **secure, efficient, and user-friendly** platform for managing banking activities such as account handling, transactions, loans, and customer services.

Implementation:-

1. Login Page

```
package ASimulatorSystem;
```

```
import java.awt.*;  
import java.awt.event.*;  
import javax.swing.*;  
import java.sql.*;
```

```
public class Login extends JFrame implements ActionListener{  
    JLabel l1,l2,l3;  
    JTextField tf1;  
    JPasswordField pf2;  
    JButton b1,b2,b3;
```

```
    Login(){  
        setTitle("AUTOMATED TELLER MACHINE");
```

```
        ImageIcon i1 = new  
        ImageIcon(ClassLoader.getResource("ASimulatorSystem/icons/logo.jpg"));  
        Image i2 = i1.getImage().getScaledInstance(100, 100, Image.SCALE_DEFAULT);  
        ImageIcon i3 = new ImageIcon(i2);
```

```
JLabel l11 = new JLabel(i3);  
l11.setBounds(70, 10, 100, 100);  
add(l11);
```

```
l1 = new JLabel("WELCOME TO ATM");  
l1.setFont(new Font("Oswald", Font.BOLD, 38));  
l1.setBounds(200,40,450,40);  
add(l1);
```

```
l2 = new JLabel("Card No:");  
l2.setFont(new Font("Raleway", Font.BOLD, 28));  
l2.setBounds(125,150,375,30);  
add(l2);
```

```
tf1 = new JTextField(15);  
tf1.setBounds(300,150,230,30);  
tf1.setFont(new Font("Arial", Font.BOLD, 14));  
add(tf1);
```

```
l3 = new JLabel("PIN:");  
l3.setFont(new Font("Raleway", Font.BOLD, 28));  
l3.setBounds(125,220,375,30);  
add(l3);
```

```
pf2 = new JPasswordField(15);  
pf2.setFont(new Font("Arial", Font.BOLD, 14));  
pf2.setBounds(300,220,230,30);  
add(pf2);
```

```
b1 = new JButton("SIGN IN");  
b1.setBackground(Color.BLACK);  
b1.setForeground(Color.WHITE);
```

```
b2 = new JButton("CLEAR");  
b2.setBackground(Color.BLACK);  
b2.setForeground(Color.WHITE);
```

```
b3 = new JButton("SIGN UP");  
b3.setBackground(Color.BLACK);  
b3.setForeground(Color.WHITE);
```

```
setLayout(null);

b1.setFont(new Font("Arial", Font.BOLD, 14));
b1.setBounds(300,300,100,30);
add(b1);

b2.setFont(new Font("Arial", Font.BOLD, 14));
b2.setBounds(430,300,100,30);
add(b2);

b3.setFont(new Font("Arial", Font.BOLD, 14));
b3.setBounds(300,350,230,30);
add(b3);

b1.addActionListener(this);
b2.addActionListener(this);
b3.addActionListener(this);

getContentPane().setBackground(Color.WHITE);

setSize(800,480);
setLocation(550,200);
setVisible(true);

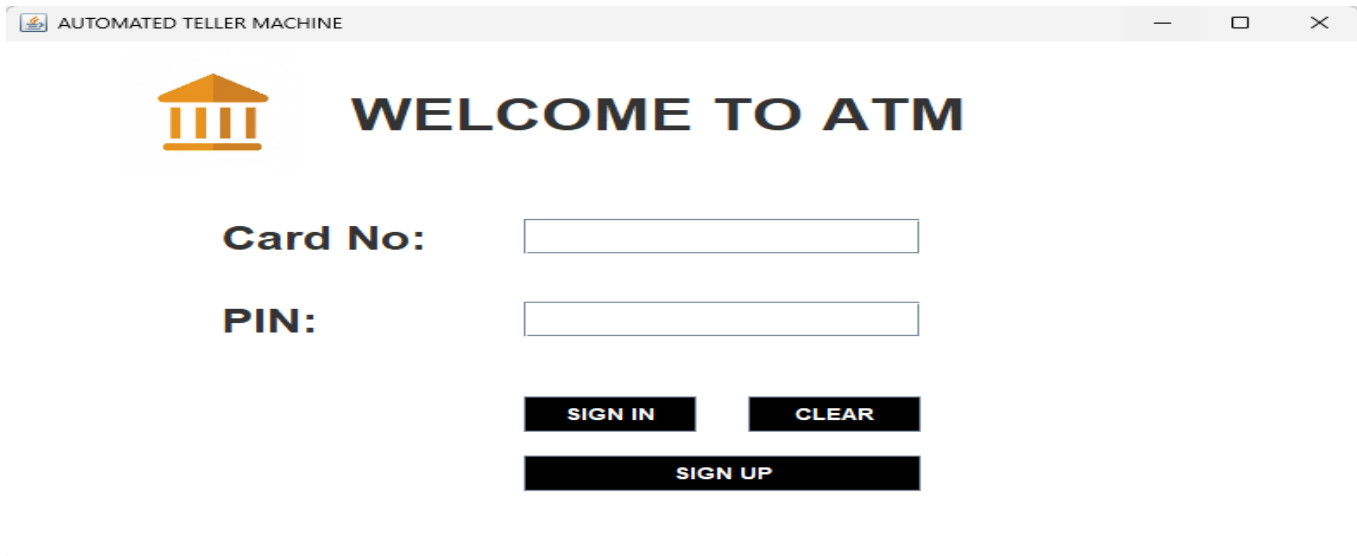
}
@Override
public void actionPerformed(ActionEvent ae){
    try{
        if(ae.getSource()==b1){
            Conn c1 = new Conn();
            String cardno = tf1.getText();
            String pin = pf2.getText();
            String q = "select * from login where cardno = '"+cardno+"' and pin = '"+pin+"'";

            ResultSet rs = (ResultSet) c1.s.executeQuery(q);
            if(rs.next()){
                setVisible(false);
                new Transactions(pin).setVisible(true);
            }else{
                JOptionPane.showMessageDialog(null, "Incorrect Card Number or PIN");
            }
        }
    }
}
```

```

    }
    }else if(ae.getSource()==b2){
        tf1.setText("");
        pf2.setText("");
    }else if(ae.getSource()==b3){
        setVisible(false);
        new Signup().setVisible(true);
    }
}catch(Exception e){
    e.printStackTrace();
}
}
public static void main(String[] args){
    new Login().setVisible(true);
}
}
}

```



WELCOME TO ATM

Card No:

PIN:

SIGN IN **CLEAR**

SIGN UP

2. Signup 1 page

```

package ASimulatorSystem;

import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import java.sql.*;
import com.toedter.calendar.JDateChooser;
import java.util.*;

```

```
public class Signup extends JFrame implements ActionListener {

    JLabel l1, l2, l3, l4, l5, l6, l7, l8, l9, l10, l11, l12, l13, l14, l15;
    JTextField t1, t2, t3, t4, t5, t6, t7;
    JRadioButton r1, r2, r3, r4, r5;
    JButton b;
    JDateChooser dateChooser;

    Random ran = new Random();
    long first4 = (ran.nextLong() % 9000L) + 1000L;
    String first = "" + Math.abs(first4);

    Signup() {

        setTitle("NEW ACCOUNT APPLICATION FORM");

        ImageIcon i1 = new
        ImageIcon(ClassLoader.getResource("ASimulatorSystem/icons/logo.jpg"));
        Image i2 = i1.getImage().getScaledInstance(100, 100, Image.SCALE_DEFAULT);
        ImageIcon i3 = new ImageIcon(i2);
        JLabel l11 = new JLabel(i3);
        l11.setBounds(20, 0, 100, 100);
        add(l11);

        l1 = new JLabel("APPLICATION FORM NO. " + first);
        l1.setFont(new Font("Raleway", Font.BOLD, 38));

        l2 = new JLabel("Page 1: Personal Details");
        l2.setFont(new Font("Raleway", Font.BOLD, 22));

        l3 = new JLabel("Name:");
        l3.setFont(new Font("Raleway", Font.BOLD, 20));

        l4 = new JLabel("Father's Name:");
        l4.setFont(new Font("Raleway", Font.BOLD, 20));

        l5 = new JLabel("Date of Birth:");
        l5.setFont(new Font("Raleway", Font.BOLD, 20));

        l6 = new JLabel("Gender:");
        l6.setFont(new Font("Raleway", Font.BOLD, 20));
```

```
l7 = new JLabel("Email Address:");  
l7.setFont(new Font("Raleway", Font.BOLD, 20));  
  
l8 = new JLabel("Marital Status:");  
l8.setFont(new Font("Raleway", Font.BOLD, 20));  
  
l9 = new JLabel("Address:");  
l9.setFont(new Font("Raleway", Font.BOLD, 20));  
  
l10 = new JLabel("City:");  
l10.setFont(new Font("Raleway", Font.BOLD, 20));  
  
l11 = new JLabel("Pin Code:");  
l11.setFont(new Font("Raleway", Font.BOLD, 20));  
  
l12 = new JLabel("State:");  
l12.setFont(new Font("Raleway", Font.BOLD, 20));  
  
l13 = new JLabel("Date");  
l13.setFont(new Font("Raleway", Font.BOLD, 14));  
  
l14 = new JLabel("Month");  
l14.setFont(new Font("Raleway", Font.BOLD, 14));  
  
l15 = new JLabel("Year");  
l15.setFont(new Font("Raleway", Font.BOLD, 14));  
  
t1 = new JTextField();  
t1.setFont(new Font("Raleway", Font.BOLD, 14));  
  
t2 = new JTextField();  
t2.setFont(new Font("Raleway", Font.BOLD, 14));  
  
t3 = new JTextField();  
t3.setFont(new Font("Raleway", Font.BOLD, 14));  
  
t4 = new JTextField();  
t4.setFont(new Font("Raleway", Font.BOLD, 14));  
  
t5 = new JTextField();
```

```
t5.setFont(new Font("Raleway", Font.BOLD, 14));

t6 = new JTextField();
t6.setFont(new Font("Raleway", Font.BOLD, 14));

t7 = new JTextField();
t7.setFont(new Font("Raleway", Font.BOLD, 14));

b = new JButton("Next");
b.setFont(new Font("Raleway", Font.BOLD, 14));
b.setBackground(Color.BLACK);
b.setForeground(Color.WHITE);

r1 = new JRadioButton("Male");
r1.setFont(new Font("Raleway", Font.BOLD, 14));
r1.setBackground(Color.WHITE);

r2 = new JRadioButton("Female");
r2.setFont(new Font("Raleway", Font.BOLD, 14));
r2.setBackground(Color.WHITE);

ButtonGroup groupgender = new ButtonGroup();
groupgender.add(r1);
groupgender.add(r2);

r3 = new JRadioButton("Married");
r3.setFont(new Font("Raleway", Font.BOLD, 14));
r3.setBackground(Color.WHITE);

r4 = new JRadioButton("Unmarried");
r4.setFont(new Font("Raleway", Font.BOLD, 14));
r4.setBackground(Color.WHITE);

r5 = new JRadioButton("Other");
r5.setFont(new Font("Raleway", Font.BOLD, 14));
r5.setBackground(Color.WHITE);

ButtonGroup groupstatus = new ButtonGroup();
groupstatus.add(r3);
groupstatus.add(r4);
groupstatus.add(r5);
```

```
dateChooser = new JDateChooser();  
//dateChooser.setBorder(new LineBorder(new Color(0, 0, 0), 1, true));  
dateChooser.setForeground(new Color(105, 105, 105));  
dateChooser.setBounds(137, 337, 200, 29);  
add(dateChooser);
```

```
setLayout(null);  
l1.setBounds(140, 20, 600, 40);  
add(l1);
```

```
l2.setBounds(290, 80, 600, 30);  
add(l2);
```

```
l3.setBounds(100, 140, 100, 30);  
add(l3);
```

```
t1.setBounds(300, 140, 400, 30);  
add(t1);
```

```
l4.setBounds(100, 190, 200, 30);  
add(l4);
```

```
t2.setBounds(300, 190, 400, 30);  
add(t2);
```

```
l5.setBounds(100, 240, 200, 30);  
add(l5);
```

```
dateChooser.setBounds(300, 240, 400, 30);
```

```
l6.setBounds(100, 290, 200, 30);  
add(l6);
```

```
r1.setBounds(300, 290, 60, 30);  
add(r1);
```

```
r2.setBounds(450, 290, 90, 30);  
add(r2);
```

```
l7.setBounds(100, 340, 200, 30);
```



```
add(l7);
```

```
t3.setBounds(300, 340, 400, 30);  
add(t3);
```

```
l8.setBounds(100, 390, 200, 30);  
add(l8);
```

```
r3.setBounds(300, 390, 100, 30);  
add(r3);
```

```
r4.setBounds(450, 390, 100, 30);  
add(r4);
```

```
r5.setBounds(635, 390, 100, 30);  
add(r5);
```

```
l9.setBounds(100, 440, 200, 30);  
add(l9);
```

```
t4.setBounds(300, 440, 400, 30);  
add(t4);
```

```
l10.setBounds(100, 490, 200, 30);  
add(l10);
```

```
t5.setBounds(300, 490, 400, 30);  
add(t5);
```

```
l11.setBounds(100, 540, 200, 30);  
add(l11);
```

```
t6.setBounds(300, 540, 400, 30);  
add(t6);
```

```
l12.setBounds(100, 590, 200, 30);  
add(l12);
```

```
t7.setBounds(300, 590, 400, 30);  
add(t7);
```

```
b.setBounds(620, 660, 80, 30);
add(b);

b.addActionListener(this);

getContentPane().setBackground(Color.WHITE);

setSize(850, 800);
setLocation(500, 120);
setVisible(true);
}

public void actionPerformed(ActionEvent ae) {

    String formno = first;
    String name = t1.getText();
    String fname = t2.getText();
    String dob = ((JTextField)
dateChooser.getDateEditor().getUiComponent()).getText();
    String gender = null;
    if (r1.isSelected()) {
        gender = "Male";
    } else if (r2.isSelected()) {
        gender = "Female";
    }
    }

    String email = t3.getText();
    String marital = null;
    if (r3.isSelected()) {
        marital = "Married";
    } else if (r4.isSelected()) {
        marital = "Unmarried";
    } else if (r5.isSelected()) {
        marital = "Other";
    }
    }

    String address = t4.getText();
    String city = t5.getText();
    String pincode = t6.getText();
    String state = t7.getText();
```

```
try {

    if (t6.getText().equals("")) {
        JOptionPane.showMessageDialog(null, "Fill all the required fields");
    } else {
        Conn c1 = new Conn();
        String q1 = "insert into signup values('" + formno + "','" + name + "','" +
fname + "','" + dob + "','" + gender + "','" + email + "','" + marital + "','" + address +
"', '" + city + "','" + pincode + "','" + state + "')";
        c1.s.executeUpdate(q1);

        new Signup2(first).setVisible(true);
        setVisible(false);
    }

} catch (Exception e) {
    e.printStackTrace();
}

}

class Conn {

    Connection c;
    Statement s;

    public Conn() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            c =
DriverManager.getConnection("jdbc:mysql://localhost:3306/bankmanagementsyste
m", "root", "bnniraj@123");
            s = c.createStatement();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

}

public static void main(String[] args) {
```

```
new Signup().setVisible(true);
```

NEW ACCOUNT APPLICATION FORM



APPLICATION FORM NO. 4927

Page 1: Personal Details

Name:	MR Niraj Kumar
Father's Name:	Sanjay Kumar Sharama
Date of Birth:	Sep 4, 2000 
Gender:	☒ Male ☐ Female
Email Address:	ndrjwhej232@gmail.com
Marital Status:	☒ Married ☐ Unmarried ☐ Other
Address:	ggg
City:	Bhagalpur
Pin Code:	812002
State:	Bihar

Next

```
}  
}
```

3.Signup 2

```
package ASimulatorSystem;
```

```
import java.awt.*;
```

```
import java.awt.event.*;
```

```
import javax.swing.*;
```

```
import java.sql.*;
```

```
public class Signup2 extends JFrame implements ActionListener{

    JLabel l1,l2,l3,l4,l5,l6,l7,l8,l9,l10,l11,l12,l13;
    JButton b;
    JRadioButton r1,r2,r3,r4;
    JTextField t1,t2,t3;
    JComboBox c1,c2,c3,c4,c5;
    String formno;
    Signup2(String formno){

        ImageIcon i1 = new
        ImageIcon(ClassLoader.getResource("ASimulatorSystem/icons/logo.jpg"));
        Image i2 = i1.getImage().getScaledInstance(100, 100, Image.SCALE_DEFAULT);
        ImageIcon i3 = new ImageIcon(i2);
        JLabel l14 = new JLabel(i3);
        l14.setBounds(150, 0, 100, 100);
        add(l14);

        this.formno = formno;
        setTitle("NEW ACCOUNT APPLICATION FORM - PAGE 2");

        l1 = new JLabel("Page 2: Additonal Details");
        l1.setFont(new Font("Raleway", Font.BOLD, 22));

        l2 = new JLabel("Religion:");
        l2.setFont(new Font("Raleway", Font.BOLD, 18));

        l3 = new JLabel("Category:");
        l3.setFont(new Font("Raleway", Font.BOLD, 18));

        l4 = new JLabel("Income:");
        l4.setFont(new Font("Raleway", Font.BOLD, 18));

        l5 = new JLabel("Educational");
        l5.setFont(new Font("Raleway", Font.BOLD, 18));

        l11 = new JLabel("Qualification:");
        l11.setFont(new Font("Raleway", Font.BOLD, 18));

        l6 = new JLabel("Occupation:");
```

```
l6.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l7 = new JLabel("PAN Number:");  
l7.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l8 = new JLabel("Aadhar Number:");  
l8.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l9 = new JLabel("Senior Citizen:");  
l9.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l10 = new JLabel("Existing Account:");  
l10.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l12 = new JLabel("Form No:");  
l12.setFont(new Font("Raleway", Font.BOLD, 13));
```

```
l13 = new JLabel(formno);  
l13.setFont(new Font("Raleway", Font.BOLD, 13));
```

```
b = new JButton("Next");  
b.setFont(new Font("Raleway", Font.BOLD, 14));  
b.setBackground(Color.BLACK);  
b.setForeground(Color.WHITE);
```

```
t1 = new JTextField();  
t1.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
t2 = new JTextField();  
t2.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
r1 = new JRadioButton("Yes");  
r1.setFont(new Font("Raleway", Font.BOLD, 14));  
r1.setBackground(Color.WHITE);
```

```
r2 = new JRadioButton("No");  
r2.setFont(new Font("Raleway", Font.BOLD, 14));  
r2.setBackground(Color.WHITE);
```

```
r3 = new JRadioButton("Yes");  
r3.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
r3.setBackground(Color.WHITE);
```

```
r4 = new JRadioButton("No");  
r4.setFont(new Font("Raleway", Font.BOLD, 14));  
r4.setBackground(Color.WHITE);
```

```
String religion[] = {"Hindu","Muslim","Sikh","Christian","Other"};  
c1 = new JComboBox(religion);  
c1.setBackground(Color.WHITE);  
c1.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
String category[] = {"General","OBC","SC","ST","Other"};  
c2 = new JComboBox(category);  
c2.setBackground(Color.WHITE);  
c2.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
String income[] = {"Null","<1,50,000","<2,50,000","<5,00,000","Upto  
10,00,000","Above 10,00,000"};  
c3 = new JComboBox(income);  
c3.setBackground(Color.WHITE);  
c3.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
String education[] = {"Non-Graduate","Graduate","Post-  
Graduate","Doctrate","Others"};  
c4 = new JComboBox(education);  
c4.setBackground(Color.WHITE);  
c4.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
String occupation[] = {"Salaried","Self-  
Employed","Business","Student","Retired","Others"};  
c5 = new JComboBox(occupation);  
c5.setBackground(Color.WHITE);  
c5.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
setLayout(null);
```

```
l12.setBounds(700,10,60,30);  
add(l12);
```

```
l13.setBounds(760,10,60,30);  
add(l13);
```

```
l1.setBounds(280,30,600,40);  
add(l1);
```

```
l2.setBounds(100,120,100,30);  
add(l2);
```

```
c1.setBounds(350,120,320,30);  
add(c1);
```

```
l3.setBounds(100,170,100,30);  
add(l3);
```

```
c2.setBounds(350,170,320,30);  
add(c2);
```

```
l4.setBounds(100,220,100,30);  
add(l4);
```

```
c3.setBounds(350,220,320,30);  
add(c3);
```

```
l5.setBounds(100,270,150,30);  
add(l5);
```

```
c4.setBounds(350,270,320,30);  
add(c4);
```

```
l11.setBounds(100,290,150,30);  
add(l11);
```

```
l6.setBounds(100,340,150,30);  
add(l6);
```

```
c5.setBounds(350,340,320,30);  
add(c5);
```

```
l7.setBounds(100,390,150,30);
```



```
add(l7);

t1.setBounds(350,390,320,30);
add(t1);

l8.setBounds(100,440,180,30);
add(l8);

t2.setBounds(350,440,320,30);
add(t2);

l9.setBounds(100,490,150,30);
add(l9);

r1.setBounds(350,490,100,30);
add(r1);

r2.setBounds(460,490,100,30);
add(r2);

l10.setBounds(100,540,180,30);
add(l10);

r3.setBounds(350,540,100,30);
add(r3);

r4.setBounds(460,540,100,30);
add(r4);

b.setBounds(570,640,100,30);
add(b);

b.addActionListener(this);

getContentPane().setBackground(Color.WHITE);

setSize(850,750);
setLocation(500,120);
setVisible(true);
}
```

```
public void actionPerformed(ActionEvent ae){
    String religion = (String)c1.getSelectedItem();
    String category = (String)c2.getSelectedItem();
    String income = (String)c3.getSelectedItem();
    String education = (String)c4.getSelectedItem();
    String occupation = (String)c5.getSelectedItem();

    String pan = t1.getText();
    String aadhar = t2.getText();

    String scitizen = "";
    if(r1.isSelected()){
        scitizen = "Yes";
    }
    else if(r2.isSelected()){
        scitizen = "No";
    }

    String eaccount = "";
    if(r3.isSelected()){
        eaccount = "Yes";
    }else if(r4.isSelected()){
        eaccount = "No";
    }

    try{
        if(t2.getText().equals("")){
            JOptionPane.showMessageDialog(null, "Fill all the required fields");
        }else{
            Conn c1 = new Conn();
            String q1 = "insert into signuptwo
values('"+formno+"','"+religion+"','"+category+"','"+income+"','"+education+"','"+occ
upation+"','"+pan+"','"+aadhar+"','"+scitizen+"','"+eaccount+"')";
            c1.s.executeUpdate(q1);

            new Signup3(formno).setVisible(true);
            setVisible(false);
        }
    }
```

```
        }catch(Exception ex){
            ex.printStackTrace();
        }

    }

    class Conn {

        Connection c;
        Statement s;

        public Conn() {
            try {
                Class.forName("com.mysql.cj.jdbc.Driver");
                c =
DriverManager.getConnection("jdbc:mysql://localhost:3306/bankmanagementsyste
m", "root", "bnniraj@123");
                s = c.createStatement();
            } catch (Exception e) {
                e.printStackTrace();
            }
        }

    }

    public static void main(String[] args){
        new Signup2("").setVisible(true);
    }
}
```



Page 2: Additional Details

Form No: 4927

Religion:	Hindu
Category:	General
Income:	Null
Educational Qualification:	Non-Graduate
Occupation:	Salaried
PAN Number:	6578r5678
Aadhar Number:	4566 6777 7789 8877
Senior Citizen:	☒ Yes ☐ No
Existing Account:	☐ Yes ☒ No

Next

4. Signup 3.

```
package ASimulatorSystem;
```

```
import java.awt.*;  
import java.awt.event.*;  
import javax.swing.*;  
import java.sql.*;  
import java.util.*;
```

```
public class Signup3 extends JFrame implements ActionListener{
```

```
    JLabel l1,l2,l3,l4,l5,l6,l7,l8,l9,l10,l11,l12;  
    JRadioButton r1,r2,r3,r4;  
    JButton b1,b2;  
    JCheckBox c1,c2,c3,c4,c5,c6,c7;  
    String formno;  
    Signup3(String formno){  
        this.formno = formno;
```

```
setTitle("NEW ACCOUNT APPLICATION FORM - PAGE 3");
```

```
ImageIcon i1 = new  
ImageIcon(ClassLoader.getResource("ASimulatorSystem/icons/logo.jpg"));  
Image i2 = i1.getImage().getScaledInstance(100, 100, Image.SCALE_DEFAULT);  
ImageIcon i3 = new ImageIcon(i2);  
JLabel l14 = new JLabel(i3);  
l14.setBounds(150, 0, 100, 100);  
add(l14);
```

```
l1 = new JLabel("Page 3: Account Details");  
l1.setFont(new Font("Raleway", Font.BOLD, 22));
```

```
l2 = new JLabel("Account Type:");  
l2.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l3 = new JLabel("Card Number:");  
l3.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l4 = new JLabel("XXXX-XXXX-XXXX-4184");  
l4.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l5 = new JLabel("(Your 16-digit Card number)");  
l5.setFont(new Font("Raleway", Font.BOLD, 12));
```

```
l6 = new JLabel("It would appear on ATM Card/Cheque Book and Statements");  
l6.setFont(new Font("Raleway", Font.BOLD, 12));
```

```
l7 = new JLabel("PIN:");  
l7.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l8 = new JLabel("XXXX");  
l8.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l9 = new JLabel("(4-digit password)");  
l9.setFont(new Font("Raleway", Font.BOLD, 12));
```

```
l10 = new JLabel("Services Required:");  
l10.setFont(new Font("Raleway", Font.BOLD, 18));
```

```
l11 = new JLabel("Form No:");
```

```
l11.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
l12 = new JLabel(formno);
```

```
l12.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
b1 = new JButton("Submit");
```

```
b1.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
b1.setBackground(Color.BLACK);
```

```
b1.setForeground(Color.WHITE);
```

```
b2 = new JButton("Cancel");
```

```
b2.setFont(new Font("Raleway", Font.BOLD, 14));
```

```
b2.setBackground(Color.BLACK);
```

```
b2.setForeground(Color.WHITE);
```

```
c1 = new JCheckBox("ATM CARD");
```

```
c1.setBackground(Color.WHITE);
```

```
c1.setFont(new Font("Raleway", Font.BOLD, 16));
```

```
c2 = new JCheckBox("Internet Banking");
```

```
c2.setBackground(Color.WHITE);
```

```
c2.setFont(new Font("Raleway", Font.BOLD, 16));
```

```
c3 = new JCheckBox("Mobile Banking");
```

```
c3.setBackground(Color.WHITE);
```

```
c3.setFont(new Font("Raleway", Font.BOLD, 16));
```

```
c4 = new JCheckBox("EMAIL Alerts");
```

```
c4.setBackground(Color.WHITE);
```

```
c4.setFont(new Font("Raleway", Font.BOLD, 16));
```

```
c5 = new JCheckBox("Cheque Book");
```

```
c5.setBackground(Color.WHITE);
```

```
c5.setFont(new Font("Raleway", Font.BOLD, 16));
```

```
c6 = new JCheckBox("E-Statement");
```

```
c6.setBackground(Color.WHITE);
```

```
c6.setFont(new Font("Raleway", Font.BOLD, 16));
```

```
c7 = new JCheckBox("I hereby declares that the above entered details correct to
```

```
th best of my knowledge.",true);
c7.setBackground(Color.WHITE);
c7.setFont(new Font("Raleway", Font.BOLD, 12));

r1 = new JRadioButton("Saving Account");
r1.setFont(new Font("Raleway", Font.BOLD, 16));
r1.setBackground(Color.WHITE);

r2 = new JRadioButton("Fixed Deposit Account");
r2.setFont(new Font("Raleway", Font.BOLD, 16));
r2.setBackground(Color.WHITE);

r3 = new JRadioButton("Current Account");
r3.setFont(new Font("Raleway", Font.BOLD, 16));
r3.setBackground(Color.WHITE);

r4 = new JRadioButton("Recurring Deposit Account");
r4.setFont(new Font("Raleway", Font.BOLD, 16));
r4.setBackground(Color.WHITE);

ButtonGroup groupgender = new ButtonGroup();
groupgender.add(r1);
groupgender.add(r2);
groupgender.add(r3);
groupgender.add(r4);

setLayout(null);

l11.setBounds(700,10,70,30);
add(l11);

l12.setBounds(770,10,40,30);
add(l12);

l1.setBounds(280,40,400,40);
add(l1);

l2.setBounds(100,140,200,30);
add(l2);
```

```
r1.setBounds(100,180,150,30);  
add(r1);
```

```
r2.setBounds(350,180,300,30);  
add(r2);
```

```
r3.setBounds(100,220,250,30);  
add(r3);
```

```
r4.setBounds(350,220,250,30);  
add(r4);
```

```
l3.setBounds(100,300,200,30);  
add(l3);
```

```
l4.setBounds(330,300,250,30);  
add(l4);
```

```
l5.setBounds(100,330,200,20);  
add(l5);
```

```
l6.setBounds(330,330,500,20);  
add(l6);
```

```
l7.setBounds(100,370,200,30);  
add(l7);
```

```
l8.setBounds(330,370,200,30);  
add(l8);
```

```
l9.setBounds(100,400,200,20);  
add(l9);
```

```
l10.setBounds(100,450,200,30);  
add(l10);
```

```
c1.setBounds(100,500,200,30);  
add(c1);
```

```
c2.setBounds(350,500,200,30);  
add(c2);
```



```
c3.setBounds(100,550,200,30);  
add(c3);
```

```
c4.setBounds(350,550,200,30);  
add(c4);
```

```
c5.setBounds(100,600,200,30);  
add(c5);
```

```
c6.setBounds(350,600,200,30);  
add(c6);
```

```
c7.setBounds(100,680,600,20);  
add(c7);
```

```
b1.setBounds(250,720,100,30);  
add(b1);
```

```
b2.setBounds(420,720,100,30);  
add(b2);
```

```
getContentPane().setBackground(Color.WHITE);
```

```
setSize(850,850);  
setLocation(500,120);  
setVisible(true);
```

```
b1.addActionListener(this);  
b2.addActionListener(this);
```

```
}
```

```
public void actionPerformed(ActionEvent ae){  
    String atype = null;  
    if(r1.isSelected()){  
        atype = "Saving Account";  
    }  
    else if(r2.isSelected()){  
        atype = "Fixed Deposit Account";  
    }  
}
```

```
}  
else if(r3.isSelected()){  
    atype = "Current Account";  
}else if(r4.isSelected()){  
    atype = "Recurring Deposit Account";  
}
```

```
Random ran = new Random();  
long first7 = (ran.nextLong() % 900000000L) + 5040936000000000L;  
String cardno = "" + Math.abs(first7);
```

```
long first3 = (ran.nextLong() % 9000L) + 1000L;  
String pin = "" + Math.abs(first3);
```

```
String facility = "";  
if(c1.isSelected()){  
    facility = facility + " ATM Card";  
}  
if(c2.isSelected()){  
    facility = facility + " Internet Banking";  
}  
if(c3.isSelected()){  
    facility = facility + " Mobile Banking";  
}  
if(c4.isSelected()){  
    facility = facility + " EMAIL Alerts";  
}  
if(c5.isSelected()){  
    facility = facility + " Cheque Book";  
}  
if(c6.isSelected()){  
    facility = facility + " E-Statement";  
}
```

```
try{  
    if(ae.getSource()==b1){  
  
        if(atype.equals("")){  
            JOptionPane.showMessageDialog(null, "Fill all the required fields");  
        }else{  
            Conn c1 = new Conn();
```

```
String q1 = "insert into signup3
values('"+formno+"','"+atype+"','"+cardno+"','"+pin+"','"+facility+"')";
String q2 = "insert into login values('"+formno+"','"+cardno+"','"+pin+"')";
c1.s.executeUpdate(q1);
c1.s.executeUpdate(q2);
JOptionPane.showMessageDialog(null, "Card Number: " + cardno + "\n
Pin:" + pin);

        new Deposit(pin).setVisible(true);
        setVisible(false);
    }

    }else if(ae.getSource()==b2){
        System.exit(0);
    }

} catch (Exception ex) {
    ex.printStackTrace();
} }

class Conn {

    Connection c;
    Statement s;

    public Conn() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            c =
DriverManager.getConnection("jdbc:mysql://localhost:3306/bankmanagementsyste
m", "root", "bnniraj@123");
            s = c.createStatement();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

public static void main(String[] args){
    new Signup3("").setVisible(true);
}
```

}
NEW ACCOUNT APPLICATION FORM - PAGE 3

Form No: 4927

 **Page 3: Account Details**

Account Type:

☐ Saving Account ☐ Fixed Deposit Account

☐ Current Account ☐ Recurring Deposit Account

Card Number: XXXX-XXXX-XXXX-4184
(Your 16-digit Card number) It would appear on ATM Card/Cheque Book and Statements

PIN: XXXX
(4-digit password)

Services Required:

☐ ATM CARD ☐ Internet Banking

☐ Mobile Banking ☐ EMAIL Alerts

☐ Cheque Book ☐ E-Statement

☒ I hereby declares that the above entered details correct to th best of my knowledge.

Submit **Cancel**

```
package ASimulatorSystem;
```

```
import java.awt.*;  
import java.awt.event.*;  
import javax.swing.*;  
import java.util.*;
```

```
public class Deposit extends JFrame implements ActionListener{
```

```
    JTextField t1,t2;
    JButton b1,b2,b3;
    JLabel l1,l2,l3;
    String pin;
    Deposit(String pin){
        this.pin = pin;
        ImageIcon i1 = new
ImageIcon(ClassLoader.getResource("ASimulatorSystem/icons/atm.jpg"));
        Image i2 = i1.getImage().getScaledInstance(1000, 1180, Image.SCALE_DEFAULT);
        ImageIcon i3 = new ImageIcon(i2);
        JLabel l3 = new JLabel(i3);
        l3.setBounds(0, 0, 960, 1080);
        add(l3);

        l1 = new JLabel("ENTER AMOUNT YOU WANT TO DEPOSIT");
        l1.setForeground(Color.WHITE);
        l1.setFont(new Font("System", Font.BOLD, 16));

        t1 = new JTextField();
        t1.setFont(new Font("Raleway", Font.BOLD, 22));

        b1 = new JButton("DEPOSIT");
        b2 = new JButton("BACK");

        setLayout(null);

        l1.setBounds(190,350,400,35);
        l3.add(l1);

        t1.setBounds(190,420,320,25);
        l3.add(t1);

        b1.setBounds(390,588,150,35);
        l3.add(b1);

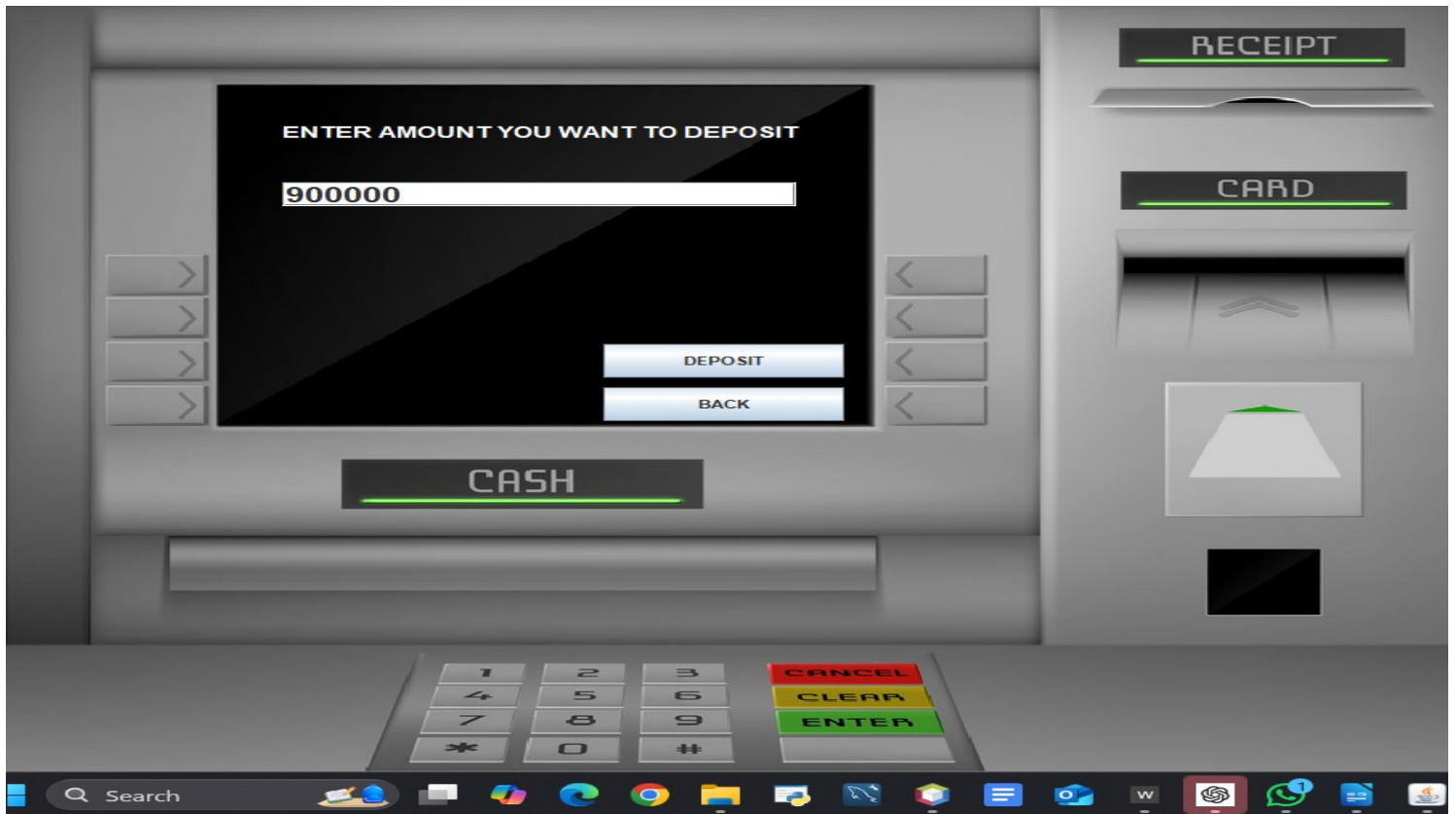
        b2.setBounds(390,633,150,35);
        l3.add(b2);

        b1.addActionListener(this);
        b2.addActionListener(this);
```

```
setSize(960,1080);
setUndecorated(true);
setLocation(500,0);
setVisible(true);
}

public void actionPerformed(ActionEvent ae){
    try{
        String amount = t1.getText();
        Date date = new Date();
        if(ae.getSource()==b1){
            if(t1.getText().equals("")){
                JOptionPane.showMessageDialog(null, "Please enter the Amount to you
want to Deposit");
            }else{
                Conn c1 = new Conn();
                c1.s.executeUpdate("insert into bank values('"+pin+"', '"+date+"',
'Deposit', '"+amount+"')");
                JOptionPane.showMessageDialog(null, "Rs. "+amount+" Deposited
Successfully");
                setVisible(false);
                new Transactions(pin).setVisible(true);
            }
        }else if(ae.getSource()==b2){
            setVisible(false);
            new Transactions(pin).setVisible(true);
        }
    }catch(Exception e){
        e.printStackTrace();
    }
}

public static void main(String[] args){
    new Deposit("").setVisible(true);
}
```



}

4. Withdroll:--

```
package ASimulatorSystem;
```

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import java.util.Date;
import java.sql.*;
```

```
public class Withdrawl extends JFrame implements ActionListener{
```

```
    JTextField t1,t2;
    JButton b1,b2,b3;
    JLabel l1,l2,l3,l4;
    String pin;
    Withdrawl(String pin){
        this.pin = pin;
        ImageIcon i1 = new
```

```
ImageIcon(ClassLoader.getResource("ASimulatorSystem/icons/atm.jpg"));
Image i2 = i1.getImage().getScaledInstance(1000, 1180, Image.SCALE_DEFAULT);
ImageIcon i3 = new ImageIcon(i2);
JLabel l3 = new JLabel(i3);
l3.setBounds(0, 0, 960, 1080);
add(l3);

l1 = new JLabel("MAXIMUM WITHDRAWAL IS RS.10,000");
l1.setForeground(Color.WHITE);
l1.setFont(new Font("System", Font.BOLD, 16));

l2 = new JLabel("PLEASE ENTER YOUR AMOUNT");
l2.setForeground(Color.WHITE);
l2.setFont(new Font("System", Font.BOLD, 16));

t1 = new JTextField();
t1.setFont(new Font("Raleway", Font.BOLD, 25));

b1 = new JButton("WITHDRAW");
b2 = new JButton("BACK");

setLayout(null);

l1.setBounds(190,350,400,20);
l3.add(l1);

l2.setBounds(190,400,400,20);
l3.add(l2);

t1.setBounds(190,450,330,30);
l3.add(t1);

b1.setBounds(390,588,150,35);
l3.add(b1);

b2.setBounds(390,633,150,35);
l3.add(b2);

b1.addActionListener(this);
b2.addActionListener(this);
```



```
setSize(960,1080);
setLocation(500,0);
setUndecorated(true);
setVisible(true);
}

public void actionPerformed(ActionEvent ae){
    try{
        String amount = t1.getText();
        Date date = new Date();
        if(ae.getSource()==b1){
            if(t1.getText().equals("")){
                JOptionPane.showMessageDialog(null, "Please enter the Amount to you
want to Withdraw");
            }else{
                Conn c1 = new Conn();

                ResultSet rs = (ResultSet) c1.s.executeQuery("select * from bank where
pin = '"+pin+"'");
                int balance = 0;
                while(rs.next()){
                    if(rs.getString("mode").equals("Deposit")){
                        balance += Integer.parseInt((String) rs.getString("amount"));
                    }else{
                        balance -= Integer.parseInt((String) rs.getString("amount"));
                    }
                }
                if(balance < Integer.parseInt(amount)){
                    JOptionPane.showMessageDialog(null, "Insuffient Balance");
                    return;
                }

                c1.s.executeUpdate("insert into bank values('"+pin+"', '"+date+"',
'Withdrawl', '"+amount+"'");
                JOptionPane.showMessageDialog(null, "Rs. "+amount+" Debited
Successfully");

                setVisible(false);
                new Transactions(pin).setVisible(true);
            }
        }
    }
}
```

```

    }else if(ae.getSource()==b2){
        setVisible(false);
        new Transactions(pin).setVisible(true);
    }
}catch(Exception e){
    e.printStackTrace();
    System.out.println("error: "+e);
}

}

public static void main(String[] args){
    new Withdrawl("").setVisible(true);
}
}

```



5. Blance Enqaury:--

```

package ASimulatorSystem;

import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

```

```
import java.util.Date;  
import java.sql.*;
```

```
public class Withdrawl extends JFrame implements ActionListener{
```

```
    JTextField t1,t2;  
    JButton b1,b2,b3;  
    JLabel l1,l2,l3,l4;  
    String pin;  
    Withdrawl(String pin){  
        this.pin = pin;  
        ImageIcon i1 = new  
ImageIcon(ClassLoader.getResource("ASimulatorSystem/icons/atm.jpg"));  
        Image i2 = i1.getImage().getScaledInstance(1000, 1180, Image.SCALE_DEFAULT);  
        ImageIcon i3 = new ImageIcon(i2);  
        JLabel l3 = new JLabel(i3);  
        l3.setBounds(0, 0, 960, 1080);  
        add(l3);  
  
        l1 = new JLabel("MAXIMUM WITHDRAWAL IS RS.10,000");  
        l1.setForeground(Color.WHITE);  
        l1.setFont(new Font("System", Font.BOLD, 16));  
  
        l2 = new JLabel("PLEASE ENTER YOUR AMOUNT");  
        l2.setForeground(Color.WHITE);  
        l2.setFont(new Font("System", Font.BOLD, 16));  
  
        t1 = new JTextField();  
        t1.setFont(new Font("Raleway", Font.BOLD, 25));  
  
        b1 = new JButton("WITHDRAW");  
        b2 = new JButton("BACK");  
  
        setLayout(null);  
  
        l1.setBounds(190,350,400,20);  
        l3.add(l1);  
  
        l2.setBounds(190,400,400,20);  
        l3.add(l2);
```

```
t1.setBounds(190,450,330,30);
l3.add(t1);

b1.setBounds(390,588,150,35);
l3.add(b1);

b2.setBounds(390,633,150,35);
l3.add(b2);

b1.addActionListener(this);
b2.addActionListener(this);

setSize(960,1080);
setLocation(500,0);
setUndecorated(true);
setVisible(true);
}

public void actionPerformed(ActionEvent ae){
    try{
        String amount = t1.getText();
        Date date = new Date();
        if(ae.getSource()==b1){
            if(t1.getText().equals("")){
                JOptionPane.showMessageDialog(null, "Please enter the Amount to you
want to Withdraw");
            }else{
                Conn c1 = new Conn();

                ResultSet rs = (ResultSet) c1.s.executeQuery("select * from bank where
pin = '"+pin+"'");
                int balance = 0;
                while(rs.next()){
                    if(rs.getString("mode").equals("Deposit")){
                        balance += Integer.parseInt((String) rs.getString("amount"));
                    }else{
                        balance -= Integer.parseInt((String) rs.getString("amount"));
                    }
                }
                if(balance < Integer.parseInt(amount)){
```

```

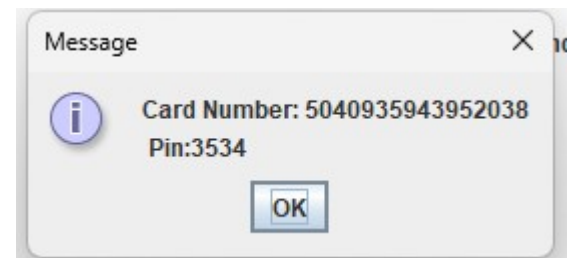
        JOptionPane.showMessageDialog(null, "Insufficient Balance");
        return;
    }

    c1.s.executeUpdate("insert into bank values('"+pin+"', '"+date+"',
'Withdrawl', '"+amount+"')");
    JOptionPane.showMessageDialog(null, "Rs. "+amount+" Debited
Successfully");

    setVisible(false);
    new Transactions(pin).setVisible(true);
}
}else if(ae.getSource()==b2){
    setVisible(false);
    new Transactions(pin).setVisible(true);
}
}catch(Exception e){
    e.printStackTrace();
    System.out.println("error: "+e);
}
}

public static void main(String[] args){
    new Withdrawl("").setVisible(true);
}
}

```



Learning Outcomes of a Bank Management System

1. Database Management:

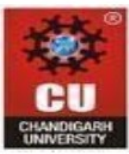
- Understanding relational databases and SQL for storing and managing banking records.
- Learning how to implement transactions and ACID properties.

2. Java EE and Backend Development:

- Using Servlets, JSP, Hibernate, and JNDI to build a dynamic banking system.
- Implementing authentication, authorization, and role-based access control.

3. Enterprise JavaBeans (EJB) & Validation:

- Learning how to create and manage enterprise beans for business logic.
- Implementing Bean Validation for data integrity.



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